# Eigen-Séta: an isogeny KEM whose trapdoor is a hidden-degree endomorphism

Specification, correctness, heuristic security analysis and a constant-time implementation

Anonymous

Revision 2 of 2026-10-17 (first draft 2026-10-11) — research draft, not peer reviewed

**Abstract**

We specify Eigen-Séta v2, a key encapsulation mechanism built on supersingular isogenies in which the decryption trapdoor is a secret endomorphism of the public curve. The public key is a curve $E_A$ together with an eigenbasis, on its $2^E$-torsion, of the endomorphism $\tau = \varepsilon\omega\hat{\varepsilon}$, where $\omega$ is the degree-2 endomorphism of the curve with complex multiplication by $\mathbb{Z}[(1+\sqrt{-7})/2]$ and $\varepsilon$ is an isogeny of *secret* degree $f$. A ciphertext is the codomain of a $3^F$-isogeny together with the images of that eigenbasis under FESTA-style diagonal masks. The masks commute with the eigen-structure, so the holder of $f$ decodes the ciphertext with a single $(2^E, 2^E)$-isogeny in dimension two (Kani's lemma), while to anyone else the ciphertext is a masked-torsion isogeny instance. We give the algorithms, prove correctness (including that the kernel used in decryption is always cyclic), derive parameters, and analyse key- and message-recovery attacks. This second revision adds an assessment. We show that a candidate secret degree can be tested on the public key alone with a Kani chain of about $2\log_2 f$ steps, half the length of a decapsulation. We implemented this test: it accepted the true degree and rejected all 158 wrong candidates at L1, L3 and L5. It lowers the cost of the degree search but leaves it far from binding. The binding attack is still guessing the mask ratio modulo about $\sqrt{3^F}$, at about $2^\lambda$ classical and $2^{\lambda/2}$ Grover queries. We therefore withdraw revision 1's conjecture of NIST categories 1, 3 and 5 for the sets L1, L3 and L5 in its unqualified form. Those sets meet the AES references only in NIST's gate count under MAXDEPTH. At least $2^\lambda$ quantum queries is met only by L5 (as the level-1 set) and by quantum-hardened sets Q3 and Q5 with 3904- and 5192-bit primes. Both claims assume that the hidden orientation of $E_A$ cannot be made effective. We identify this as the scheme's largest risk, because an effective orientation would give subexponential quantum key recovery at every set. At level 1, public keys and ciphertexts are 1344 bytes. The maintained Rust implementation decapsulates with a theta chain in about 1.3 s on a 4-core cloud VM. The security of the scheme is a conjecture supported by the analysis in this paper only; it has not been reviewed by others.

## Contents

**1 Introduction** **2**

**2 Preliminaries** **3**

**3 The scheme** **4**
3.1 Parameters . . . . . 4
3.2 Key generation . . . . . 4
3.3 Encapsulation . . . . . 5
3.4 Decapsulation . . . . . 5
3.5 The transform . . . . . 6

**4 Correctness** **6**

**5 Parameters** **8**

**6 Security analysis** **8**
6.1 Problems . . . 9
6.2 The public-degree variant is insecure . . . 9
6.3 Key recovery . . . 9
6.4 Message recovery . . . 11
6.5 Quantum security: two claim tiers . . . 12
6.6 Chosen-ciphertext security and decryption failures . . . 12
6.7 Weak keys . . . 13

**7 Implementation and side channels** **14**
7.1 Arithmetic . . . 14
7.2 Constant-time decapsulation . . . 14
7.3 Residual issues . . . 15
7.4 Tests . . . 15
7.5 The maintained implementation (revision 2) . . . 15

**8 Measurements** **16**
8.1 Sizes and speed . . . 16
8.2 Correctness . . . 17
8.3 Constant-time evidence . . . 17
8.4 Revision 2: the public-key degree test . . . 18

**9 Assessment (revision 2)** **19**

**10 Open problems** **20**

# 1 Introduction

After the 2022 attacks on SIDH [CD23, MMP+23, Rob23], isogeny-based public-key encryption needs a trapdoor that does not reveal enough torsion information to let the attacker run those same attacks. Two families survive. In the first, torsion images are masked so that the attacker cannot recover them exactly, as in M-SIDH [FMP23] and FESTA [BMP23]; in the second, the scheme avoids torsion images altogether. FESTA decrypts by running the torsion attack itself with the help of a secret: the legitimate receiver knows a matrix that commutes with the masks.

Castryck and Vercauteren [CV23] observed that a torsion basis which is an eigenbasis of a *known* endomorphism of small degree makes masked-torsion instances easy: the masks act diagonally on the eigenvectors, so the endomorphism's action on the masked points is known exactly, and a lollipop endomorphism can be decoded with Kani's lemma. They presented this as an attack, and as a backdoor available to whoever chose the basis.

**This work.** Eigen-Séta turns that observation into a trapdoor. The public key is a curve $E_A$ with an eigenbasis of an endomorphism $\tau \in \mathrm{End}(E_A)$ whose degree $2f^2$ is *secret*. The sender masks the images of the eigenbasis under a $3^F$-isogeny $\varphi$ with a random diagonal matrix. The receiver, who knows $f$, knows how $\theta = \varphi\tau\hat{\varphi}$ acts on the masked points and recovers $\varphi$ from one $(2^E, 2^E)$-isogeny between products of elliptic curves. Without $f$, the ciphertext is a FESTA-type masked-torsion instance, and the public key is an isogeny instance of unknown degree from a known curve.

An earlier variant [Ano26b] took $\tau = \varepsilon\omega\hat{\varepsilon}$ with $\varepsilon$ of *public* degree $5^k$. That variant is insecure: its decryption key is a function of the public degree alone (§6.2). Version 2, specified here, draws $f$ from an interval of about $2^{277}$, $2^{408}$ or $2^{535}$ values, depending on the level, and keeps it secret.

**Contributions.**

- A complete specification of the KEM (§3) with the Fujisaki–Okamoto transform with implicit rejection [HHK17].
- Proofs that decryption is correct (§4). These include a proposition that the kernel recovered during decryption is always cyclic, which settles a corner case reported for the earlier toy prototype.
- Parameter sets for $\lambda = 128, 192, 256$ (§5) and a heuristic analysis of key recovery, message recovery, quantum attacks, chosen-ciphertext attacks and decryption failures (§6). The concrete figures are summarised in Table 3.
- A Rust implementation with constant-time decapsulation, checked with an instruction-count probe and a statistical timing test. The probe found one compiler-introduced branch, which we fixed (§7).
- Measured sizes, timings and correctness rates at all parameter sets, and constant-time evidence at Toy16 and L1 (§8), each traceable to a committed record.
- Revision 2 adds an assessment (§9). It contains a public-key-only test for a candidate degree with a chain of half the decapsulation length (Lemma 8), implemented and measured (§8.4). It also contains a corrected quantum claim in two tiers (§6.5), the quantum-hardened parameter sets, and an account of the hidden orientation of $E_A$ as the main structural risk.

**What this paper does not claim.** Security is a conjecture backed by the attacks we considered; no third party has analysed the scheme. In particular, L1 and L3 are not claimed to resist $2^{128}$ or $2^{192}$ quantum queries. The implementation is research code. Revision 1's timings were measured on a Mumford-coordinate decapsulation chain. The maintained implementation [Ano26a] uses a theta-coordinate chain [DMPR24]. Key generation is not constant time.

**Evidence conventions.** Each measured number carries the path of the record it comes from, relative to the root of the repository (`crypto:`) or of the sibling repository that holds the maintained implementation (`eigenx:`). Derived quantities are marked as such.

## 2 Preliminaries

**Curves and isogenies.** Let $p$ be a prime and $\mathbb{F}_{p^2} = \mathbb{F}_p(i)$ with $i^2 = -1$ ($p \equiv 3 \bmod 4$). All curves are supersingular and defined over $\mathbb{F}_{p^2}$. Isogenies are separable. For an isogeny $\phi$ we write $\hat{\phi}$ for its dual, so that $\hat{\phi}\phi = [\deg \phi]$. For $N$ coprime to $p$, the $N$-torsion $E[N] \cong (\mathbb{Z}/N)^2$ carries the Weil pairing $e_N$, and $e_N(\phi P, \phi Q) = e_N(P, Q)^{\deg \phi}$.

**The base curve.** Let $E_D$ be the curve with $j$-invariant $-3375$. It has complex multiplication by $\mathcal{O}_K = \mathbb{Z}[(1 + \sqrt{-7})/2]$, and it is supersingular over $\mathbb{F}_p$ whenever $-7$ is not a square modulo $p$, i.e. $p \equiv 3, 5, 6 \pmod 7$. Its endomorphism $\omega$ of degree 2 satisfies $\omega^2 - t\omega + 2 = 0$ with $t = \pm 1$, so $(2\omega - t)^2 = -7$. We use the short Weierstrass model $y^2 = x^3 - 3jkx - 2jk^2$ with $j = -3375$ and $k = j - 1728$. Since $p + 1 = 2^{E+1}3^F c$, every curve in the isogeny class has $\#E(\mathbb{F}_{p^2}) = (p+1)^2$ and full rational $2^E$- and $3^F$-torsion.

**Eigenbases.** Let $s_D$ be a square root of $-7$ modulo $2^{E+2}$ (it exists because $-7 \equiv 1 \bmod 8$), and set $\mu = (t + s_D)/2$ and $\nu = (t - s_D)/2$ modulo $2^E$. These are the two roots of $x^2 - tx + 2$ modulo $2^E$, and $\mu - \nu = s_D$ is odd. Hence $\omega$ has eigenvectors $P_D, Q_D$ spanning $E_D[2^E]$ with $\omega P_D = \mu P_D$ and $\omega Q_D = \nu Q_D$.

**Kani's lemma in the form used here.** Let $E$ be a curve, $\gamma \in \mathrm{End}(E)$ and $c' \in \mathbb{Z}$ with $\deg\gamma + c'^2 = 2^E$. Then

$$\Phi = \begin{pmatrix} c' & -\hat{\gamma} \\ \gamma & c' \end{pmatrix} : E^2 \to E^2$$

satisfies $\hat{\Phi}\Phi = [2^E]$. It is therefore a $(2^E, 2^E)$-isogeny between principally polarised abelian surfaces, and its kernel is $\{(c'R, -\gamma R) : R \in E[2^E]\}$. Given $\gamma(R)$ for a basis of $E[2^E]$, the isogeny $\Phi$ can be computed as a chain of $E$ $(2,2)$-isogenies [Kan97, CD23, Rob23], and it can then be evaluated at points outside the $2^E$-torsion. In particular $\Phi(X, 0) = (c'X, \gamma X)$ reveals $\gamma$ on any subgroup.

**$(2,2)$-chains.** A chain from $E \times E$ can pass through products of elliptic curves (product steps), glue into the Jacobian of a genus-2 curve, continue with Richelot isogenies, and split back into a product. We use the gluing, Richelot and splitting formulas of Castryck and Decru [CD23] in Mumford coordinates. The order in which these step types occur is the *schedule* of the chain.

**FESTA masking.** In FESTA [BMP23], the images of a torsion basis under a secret isogeny are published multiplied by a secret matrix. We use diagonal masks: $(a\varphi P, b\varphi Q)$ with odd $a, b$ modulo $2^E$. The Weil pairing reveals $ab$. The ratio $b/a$ stays hidden.

# 3 The scheme

## 3.1 Parameters

A parameter set consists of the following data.

- Exponents $E$ and $F$ and a cofactor $c$ such that $p = 2^{E+1}3^F c - 1$ is prime and $p \equiv 3, 5, 6 \pmod 7$.
- $d = 3^F$, the sender's isogeny degree.
- A lower bound $f_{\mathrm{lo}}$ for the secret degree, which ranges over $[f_{\mathrm{lo}}, 2f_{\mathrm{lo}})$.
- The curve $E_D$, its endomorphism $\omega$ with trace $t$, the values $s_D, \mu, \nu$ and the eigenbasis $(P_D, Q_D)$ of §2.
- Domain-separated instances of SHAKE256: $G$ (masks), $H_P$ (public-key hash), $H_K$ (shared secret) and $H_R$ (rejection key).

The concrete sets are in §5.

**Canonical bases.** $\mathsf{Basis}_3(A)$ deterministically computes a basis $(P_3, Q_3)$ of $E[3^F]$ on the Montgomery curve $E : y^2 = x^3 + Ax^2 + x$. It tries $x = 1 + i, 2 + i, \ldots$, lifts each to a point, multiplies by $(p+1)/d$, and keeps the first two points whose $3^{F-1}$-multiples generate $E[3]$. It depends only on $A$, so sender and receiver agree on it.

## 3.2 Key generation

The degree-$f$ factor $\varepsilon$ exists by Kani's lemma applied to $\alpha$, since $\deg\alpha + f^2 = f \cdot 2^E$ and $\gcd(f, g) = 1$ (see [Rob23, §2]). The requirement $(y_0, y_1) \neq (0, 0)$ excludes $\alpha \in \mathbb{Z}[\omega]$, for which $E_A$ would lie in the small orbit of $E_D$ under the class group of $\mathcal{O}_K$ (§6.7). The endomorphism the public key is built around is

$$\tau = \varepsilon\omega\hat{\varepsilon} \in \mathrm{End}(E_A), \qquad \deg\tau = 2f^2, \quad \mathrm{Tr}\,\tau = tf,$$

**Algorithm 1** KeyGen()

**repeat** ▷ secret degree and Kani scalars
  $f \leftarrow$ uniform odd integer in $[f_{\mathrm{lo}}, 2f_{\mathrm{lo}})$
  reject if $3 \mid f$, $7 \mid f$ or $2f \equiv 2^E \pmod d$
  $m \leftarrow 2^E - 7d^2f^2$; find $c, c'$ with $c^2 + c'^2 = m$, $c$ even, or reject
  $s_0 \leftarrow f s_D \bmod 2^E$
  choose signs of $c, c'$ with $\max(v_2(c' - \lambda_S), v_2(c' + \lambda_S)) = 2$, where $\lambda_S = ds_0 + c$, or reject
**until** accepted
$g \leftarrow 2^E - f$
**repeat** ▷ endomorphism of norm $fg$ that uses Frobenius
  pick $(y_0, y_1) \neq (0,0)$ with $T \leftarrow fg - p\,N(y_0 + y_1\omega) > 0$ prime, $T \equiv 1, 2, 4 \pmod 7$
  solve $x^2 + 7y'^2 = T$ (Cornacchia); $x_0 \leftarrow x - ty'$, $x_1 \leftarrow 2y'$
**until** found
$\alpha \leftarrow x_0 + x_1\omega + (y_0 + y_1\omega)\pi$ ▷ $\pi$ the $p$-Frobenius; $\operatorname{nrd}\alpha = fg$
compute the $(2^E, 2^E)$-isogeny with kernel $\{([f]R, \alpha R) : R \in E_D[2^E]\}$; reject unless its first step is a gluing
from its codomain $E_1 \times E_2$, pick the factor $E_A$ whose map $\varepsilon : E_D \to E_A$ has degree $f$ (test $e(\varepsilon X, \varepsilon Y) = e(X,Y)^f$ on $E_D[3^F]$)
$\lambda_1, \lambda_2 \leftarrow$ uniform odd residues modulo $2^E$
$P_A \leftarrow \lambda_1\varepsilon(P_D)$, $Q_A \leftarrow \lambda_2\varepsilon(Q_D)$; bring $E_A$ to Montgomery form $y^2 = x^3 + Ax^2 + x$
$\lambda_T \leftarrow c - ds_0$; reject if $\lambda_S, \lambda_T$ or $c' \equiv -1 \pmod{2^E}$, or $c$ or $c' \equiv -1 \pmod d$
$pk \leftarrow (A, x(P_A), x(Q_A), x(P_A - Q_A))$; $z \leftarrow$ 32 random bytes
$sk \leftarrow (\lambda_S, \lambda_T, c' \bmod 2^E, c \bmod d, c' \bmod d, z, pk)$
**return** $(pk, sk)$ after a self-test encapsulation and decapsulation

and since $\tau\varepsilon = f\,\varepsilon\omega$, the points $P_A, Q_A$ are eigenvectors of $\tau$ on $E_A[2^E]$ with eigenvalues $f\mu$ and $f\nu$. The scalars $\lambda_1, \lambda_2$ make the Weil pairing $e(P_A, Q_A) = e(P_D, Q_D)^{\lambda_1\lambda_2 f}$ uniform among values of the same order, so it carries no information on $f$.

### 3.3 Encapsulation

**Algorithm 2** Encaps($pk$) and the deterministic core Enc($pk, s$)

$s \leftarrow$ uniform in $\mathbb{Z}/d$; $ct \leftarrow \mathsf{Enc}(pk, s)$; **return** $(ct,\ H_K(s, ct))$

**function** Enc($pk = (A, x_P, x_Q, x_{P-Q}), s$)
  $(P_3, Q_3) \leftarrow \mathsf{Basis}_3(A)$; $K \leftarrow P_3 + sQ_3$ ▷ 3-point ladder
  $\varphi : E_A \to E_C = E_A/\langle K \rangle$ ▷ chain of $F$ 3-isogenies, fixed strategy
  push $x_P, x_Q, x_{P-Q}$ through $\varphi$: $P' = \varphi(P_A)$, $Q' = \varphi(Q_A)$
  $(a, b) \leftarrow G(s, H_P(pk))$, odd residues modulo $2^E$
  **return** $(A_C,\ x(aP'),\ x(bQ'),\ x(aP' - bQ'))$
**end function**

### 3.4 Decapsulation

Write $S = aP'$ and $T = bQ'$, and $\theta = \varphi\tau\hat{\varphi} \in \mathrm{End}(E_C)$. Section 4 shows that $\psi = 2\theta - tdf$ has degree $7d^2f^2$ and trace 0, and acts on $S$ and $T$ by $\pm ds_0$. Decapsulation applies Kani's lemma to $\gamma = \psi + c$, whose degree is $7d^2f^2 + c^2 = 2^E - c'^2$.

All steps after line 2 run in constant time with respect to $sk$ and $s$ (§7). Lines 1–2 depend only on public data.

**Algorithm 3** $\mathsf{Decaps}(sk, ct)$

parse $ct = (A_C, x_S, x_T, x_{S-T})$; lift $S, T$ on $E_C$ with $x(S-T) = x_{S-T}$; on failure **return** $H_R(z, ct)$ ▷ public
$(X_1, X_2) \leftarrow \mathsf{Basis}_3(A_C)$; $(P_3, Q_3) \leftarrow \mathsf{Basis}_3(A)$ ▷ public
$\Phi \leftarrow$ the $(2^E, 2^E)$-isogeny of $E_C^2$ with kernel $\langle (c'S, -\lambda_S S), (c'T, -\lambda_T T)\rangle$, fixed schedule
evaluate $\Phi$ at $(X_1, 0), (0, X_1), (X_2, 0)$; identify both codomain factors with $E_C$ among the 8 sign and order choices by comparing with $c'X_i$; read off $\gamma(X_1), \gamma(X_2)$
$\psi(X_i) \leftarrow \gamma(X_i) - cX_i$ for $i = 1, 2$
$G \leftarrow$ whichever of $\psi(X_1), \psi(X_2)$ has order $d$; $H \leftarrow$ the other; find $k$ with $H = kG$ ▷ pairings, DLP in $\mu_d$
$\hat{\varphi} \leftarrow E_C/\langle X_h - kX_g\rangle \to E_A$; $K_g \leftarrow$ the image of $X_1$ or $X_2$ that has order $d$
$s \leftarrow \log(e(P_3, K_g))/\log(e(K_g, Q_3))$ in $\mu_d$
$ct' \leftarrow \mathsf{Enc}(pk, s)$
**return** $H_K(s, ct)$ if $ct' = ct$, else $H_R(z, ct)$ ▷ constant-time selection

### 3.5 The transform

$\mathsf{Enc}$ is a deterministic public-key encryption of the message $s \in \mathbb{Z}/d$, derandomised by deriving the masks from $G(s, H_P(pk))$. The KEM is the $U^{\not\perp}$ transform with implicit rejection of Hofheinz, Hövelmanns and Kiltz [HHK17] applied to it. The re-encryption check compares the entire ciphertext, including the curve coefficient.

## 4 Correctness

Throughout, $(pk, sk)$ is honestly generated, $ct = \mathsf{Enc}(pk, s)$, and the notation of Algorithms 1–3 is in force. Write $\theta = \varphi\tau\hat{\varphi} \in \mathrm{End}(E_C)$ and $\psi = 2\theta - tdf$.

**Lemma 1** (the lollipop endomorphism)**.** $\operatorname{Tr}\psi = 0$ *and* $\deg\psi = 7d^2f^2$*. Moreover* $\psi S = ds_0 S$ *and* $\psi T = -ds_0 T$*, where* $s_0 = fs_D \bmod 2^E$*.*

*Proof.* $\operatorname{Tr}\theta = \varphi(\tau + \hat{\tau})\hat{\varphi} = \varphi[tf]\hat{\varphi} = tdf$, and $\deg\theta = d^2\deg\tau = 2d^2f^2$. Hence $\operatorname{Tr}\psi = 2\operatorname{Tr}\theta - 2tdf = 0$ and

$$\deg\psi = (2\theta - tdf)(2\hat{\theta} - tdf) = 4\deg\theta - 2tdf\operatorname{Tr}\theta + t^2d^2f^2 = 8d^2f^2 - 2d^2f^2 + d^2f^2 = 7d^2f^2.$$

For the eigenvalues, $\hat{\varphi}S = a\hat{\varphi}\varphi P_A = adP_A$, and $\tau P_A = f\mu P_A$, so $\theta S = adf\mu\,\varphi P_A = df\mu S$. Likewise $\theta T = df\nu T$. Then $\psi S = df(2\mu - t)S = dfs_D S$, and $\psi T = df(2\nu - t)T = -dfs_D T$. □

The masks $a, b$ drop out because $S$ and $T$ are multiples of eigenvectors. This is where the trapdoor is used: only $f$ (through $s_0$) and the Kani scalars are needed, never $\varphi$ or the masks.

**Proposition 2** (the decryption isogeny)**.** *Let* $\gamma = \psi + c$*. Then* $\deg\gamma + c'^2 = 2^E$*, and the subgroup* $\langle (c'S, -\lambda_S S), (c'T, -\lambda_T T)\rangle$ *of* $E_C^2$ *is the kernel of the* $(2^E, 2^E)$*-isogeny* $\Phi = \left(\begin{smallmatrix} c' & -\hat{\gamma} \\ \gamma & c' \end{smallmatrix}\right)$.

*Proof.* $\deg\gamma = \deg\psi + c\operatorname{Tr}\psi + c^2 = 7d^2f^2 + c^2 = 2^E - c'^2$ by the choice of $c, c'$. By Lemma 1, $\gamma S = (ds_0 + c)S = \lambda_S S$ and $\gamma T = (c - ds_0)T = \lambda_T T$. The points $S, T$ form a basis of $E_C[2^E]$, because $\varphi$ has odd degree and $a, b$ are odd. So the two generators are $(c'R, -\gamma R)$ for $R = S, T$, and the kernel statement is Kani's lemma (§2). □

From $\Phi(X, 0) = (c'X, \gamma X)$, decapsulation obtains $\gamma$, and hence $\psi = \gamma - c$, on $E_C[d]$. Restricted to $E_C[d]$ we have $\psi = 2\varphi\tau\hat{\varphi}$, since $d \mid tdf$.

**Proposition 3** (the recovered kernel is always cyclic)**.** $\psi(E_C[d])$ *is cyclic of order* $d$*, and the kernel of* $\psi$ *restricted to* $E_C[d]$ *is exactly* $\ker\hat{\varphi}$*.*

*Proof.* We have $\hat{\varphi}(E_C[d]) = \ker\varphi = \langle K\rangle$, which is cyclic of order $d$. The endomorphism $\tau$ has degree $2f^2$, which is prime to 3, so $\tau\langle K\rangle$ is again cyclic of order $d$. The map $\varphi$ is injective on $\tau\langle K\rangle$ if and only if $\tau\langle K\rangle \cap \langle K\rangle = 0$, i.e. if and only if the two subgroups have different subgroups of order 3. That fails only if the line $L = \langle 3^{F-1}K\rangle \subset E_A[3]$ is an eigenline of $\tau$ modulo 3. The characteristic polynomial of $\tau$ is $x^2 - tfx + 2f^2$, whose discriminant is $f^2(t^2-8) = -7f^2 \equiv 2f^2 \pmod 3$. Since $3 \nmid f$, this is a non-square modulo 3. So $\tau$ has no eigenline in $E_A[3]$, and $\varphi$ maps $\tau\langle K\rangle$ isomorphically onto a cyclic group of order $d$. Thus $\psi(E_C[d]) = 2\varphi\tau\langle K\rangle$ has order $d$, its kernel has order $d^2/d = d$, and that kernel contains $\ker\hat{\varphi}$, which also has order $d$. □

*Remark* 4. The design note [Ano26b] reported a "non-cyclic $\psi'$" corner case in one of eight trials of its toy prototype, a variant with a different $\psi'$ and very small $d = 3^5$. Proposition 3 shows that the case cannot occur in version 2. The argument needs only that $-7$ is a non-square modulo 3 and $3 \nmid \deg\tau$. The cause of the toy observation was not isolated.

**Proposition 5** (recovery of $s$)**.** *Let $G$ be a generator of $\psi(E_C[d])$ and $H = kG$ the image of the other basis point. Then $\ker\hat{\varphi} = \langle X_h - kX_g\rangle$. If $K_g$ is the image under $\hat{\varphi}$ of a basis point and has order $d$, then $K_g = u(P_3 + sQ_3)$ for a unit $u$, and $s = \log e(P_3, K_g)/\log e(K_g, Q_3)$ in $\mu_d$.*

*Proof.* $\psi(X_h - kX_g) = H - kG = 0$, and $X_h - kX_g$ has order $d$, so by Proposition 3 it generates $\ker\hat{\varphi}$. The image of $E_C[d]$ under $\hat{\varphi}$ is $\ker\varphi = \langle P_3 + sQ_3\rangle$. With $\zeta = e(P_3, Q_3)$, which has order $d$, we get $e(K_g, Q_3) = \zeta^u$ and $e(P_3, K_g) = \zeta^{us}$. The codomain of $\hat{\varphi}$ is $E_A$ up to isomorphism, and isomorphisms preserve these subgroups. The implementation maps the codomain to the model with coefficient $A$, using $x \mapsto -x$ when the computed coefficient is $-A$. □

**Proposition 6** (fixed schedule)**.** *In the basis $(S,0),(T,0),(0,S),(0,T)$ of $E_C^2[2^E]$, the coordinates of $\ker\Phi$ are the columns of $\begin{pmatrix} c' & 0 \\ 0 & c' \\ -\lambda_S & 0 \\ 0 & -\lambda_T \end{pmatrix}$ modulo $2^E$, which depend on sk only. Whether each $(2,2)$-step of the chain is a product step, a gluing, a Richelot step or a splitting is decided by linear conditions on these coordinates modulo powers of 2. So the schedule is the same for every ciphertext under a given key.*

*Proof sketch.* The kernel of the $j$-th step is $2^{E-j}$ times the image of $\ker\Phi$ under the first $j-1$ steps. On a product $E_1 \times E_2$, the step is a product step exactly when its 2-torsion kernel is a direct product $\langle (P,0),(0,Q)\rangle$, and a graph step when it is the graph of the identity of $E_C[2]$ (on $E_C \times E_C$) or of an isomorphism (after the split). Both are statements about the coordinates. On a Jacobian, the step is a splitting exactly when the codomain is a product, which happens at a fixed index determined by the same 2-adic data. Key generation imposes $\max(v_2(c' \mp \lambda_S)) = 2$ and the gluing test of line 15, which fixes the schedule as graph, product, glue, Richelot$^{E-6}$, split, product, graph. The self-test checks it once. □

**Correctness error.** By the propositions above, decapsulation of an honest ciphertext returns $H_K(s, ct)$ unless an intermediate formula meets a degenerate input. Examples are a Mumford divisor not in general position, a zero denominator in a Richelot or splitting formula, or a Miller line through the second argument. No branch handles these cases. The output is then wrong but computed in the same time, and the FO check rejects it. Heuristically each such event has probability $O(1/p)$ per step, giving $\delta = O(E/p) \approx 2^{-1330}$ at level 1. Key generation excludes the one structured degenerate case, a scalar $\equiv -1$ modulo the point order, which the x-only ladder with $y$-recovery cannot handle. The measured rates are in Table 6: no failure occurred in 3,635 decapsulations.

# 5 Parameters

The parameter sets are produced by `eigenx:examples/gen_params.rs`, which writes `eigenx:src/params_generated.rs` (the historical copy is `crypto:eigen_seta/`). They satisfy four constraints.

(P1) $d = 3^F \geq 2^{m\lambda}$, with $F = \lceil m\lambda / \log_2 3 \rceil$, where $m = 2$ for the L sets and $m = 4$ for the quantum-hardened Q sets. Message recovery costs about $\sqrt{d}$ classically and $d^{1/4}$ Grover queries (§6.4, §6.5).

(P2) $2^E > 28\, d^2 f_{\max}^2$ with $f_{\max} = 2f_{\mathrm{lo}}$. Then $m = 2^E - 7d^2f^2 > \frac{3}{4}2^E$ for every admissible $f$, so the two-squares step has a large target. $E$ is the smallest exponent that works.

(P3) $f_{\mathrm{lo}} = 2^\kappa \lfloor p/2^E \rfloor + 1$. Then $fg \gtrsim 2^\kappa p$, and the norm equation $N(x) + pN(y) = fg$ has about $2^\kappa$ candidate values of $y \neq 0$. We take $\kappa = 12$ for the real sets and $\kappa = 8$ for the toys.

(P4) $p = 2^{E+1}3^F c - 1$ is prime with $p \equiv 3, 5, 6 \pmod 7$, and $c$ is the smallest cofactor that achieves this. Primality is decided by trial division and Miller–Rabin with the twelve fixed bases $2, \ldots, 37$, so the search uses no randomness and anyone can rerun it (`eigenx:docs/NUMS.md`, `eigenx:examples/verify_nums.rs`).

Since $f \approx 2^{\kappa+1}cd$ and $2^E \approx 28d^2f^2$, we get $p \approx 2^{E+1}d \approx d^5 \cdot 2^{2\kappa+O(1)}c^2$. That is $\log_2 p \approx 10\lambda$. The factor $d^5$ is the price of decrypting with a single chain, whose torsion order must exceed the degree $7d^2f^2 + c^2$ of $\gamma$. Evaluating $\Phi$ from two half-chains, as in SQIsign 2D-style implementations [DMPR24], would let $2^E$ shrink to about $\sqrt{\deg \gamma}$. We have not implemented this.

Table 1: Parameter sets. $d = 3^F$; "$f$ bits" is the bit length of the secret degree in the recorded keys; public key and ciphertext are four $\mathbb{F}_{p^2}$ elements each. Source: `eigenx:src/params_generated.rs` (L sets identical to `crypto:eigen_seta/src/params_generated.rs`); $f$ bits from `crypto:research/eigen_seta_rust_20261011/reports/run-20261011T1813Z-bench/records/bench_output.txt`.

| set | $\lambda$ | $\log_2 p$ | $E$ | $F$ | $\log_2 d$ | $c$ | $\kappa$ | $f$ bits | pk = ct (B) |
|---|---|---|---|---|---|---|---|---|---|
| Toy16 | 16 | 216 | 174 | 21 | 33.3 | 158 | 8 | 51 | 216 |
| Toy32 | 32 | 369 | 297 | 41 | 65.0 | 55 | 8 | 80 | 376 |
| L1 | 128 | 1341 | 1076 | 162 | 256.8 | 144 | 12 | 278 | 1344 |
| L3 | 192 | 1991 | 1594 | 243 | 385.1 | 1059 | 12 | 409 | 1992 |
| L5 | 256 | 2630 | 2106 | 324 | 513.5 | 380 | 12 | 536 | 2632 |
| Q3 | 192 | 3904 | 3125 | 485 | 768.7 | 371 | 12 | 791 | 3904 |
| Q5 | 256 | 5192 | 4155 | 647 | 1025.5 | 975 | 12 | 1049 | 5192 |

The toy sets have no security. They were used in the historical measurements of §8 and are no longer shipped by the maintained implementation. Rule (P1) with $m = 4$ at $\lambda = 128$ gives $F = 324$, the L5 exponent, and the rest of the search then reproduces the L5 prime. So there is no separate Q1 set: L5 is both the classical level-5 set and the quantum-hardened level-1 set. The Q rows come from `eigenx:src/params_generated.rs` and `eigenx:README.md`. Their $f$ bits are those of $f_{\mathrm{lo}}$, not of a recorded key.

# 6 Security analysis

This section is heuristic. We state the problems the scheme relies on, reduce key recovery to finding $f$, and estimate the cost of every attack we know of. Table 3 summarises the estimates.

A cost written $2^x \cdot C$ means $2^x$ evaluations of a polynomial-time subroutine $C$, typically a $(2,2)$-chain in dimension 2 or an isogeny chain in dimension 8.

### 6.1 Problems

**Assumption 1** (hidden-degree eigenbasis problem)**.** Given the public parameters and $pk = (E_A, P_A, Q_A)$, it is hard to find $f$, or any secret key that decapsulates honest ciphertexts for $pk$.

**Assumption 2** (masked eigen-torsion problem)**.** Given $pk$ and $ct = \mathsf{Enc}(pk, s)$ for uniform $s \in \mathbb{Z}/d$, it is hard to find $s$. This is the one-wayness of the deterministic PKE inside the KEM.

**Lemma 7** (the degree is the key)**.** *Given $f$, a working secret key can be computed in polynomial time, up to a choice among the few representations of $m = 2^E - 7d^2f^2$ as a sum of two squares.*

*Proof.* $s_0 = fs_D$ is public once $f$ is known. The pair $(c, c')$ is one of the representations $c^2 + c'^2 = m$; key generation finds one with the same procedure an attacker would use, and the schedule condition and the signs leave at most a handful of candidates. Each candidate can be tested by decapsulating a self-made ciphertext. Neither $\alpha$, $\varepsilon$ nor $\lambda_1, \lambda_2$ is needed. □

So Assumption 1 is equivalent to the hardness of finding $f$. A consequence is that $E_A$ and the eigen-structure must hide $f$; the next subsection shows what goes wrong when they do not.

### 6.2 The public-degree variant is insecure

The toy prototype of [Ano26b] had a "walk" variant: $E_A$ was the codomain of a secret isogeny $\varepsilon$ of *public* degree $5^k$, and $\tau = \varepsilon\omega\hat{\varepsilon}$. Its decryption key $(c, c', s_0)$ is computed from $\Delta' = 4\deg\tau - (\operatorname{Tr}\tau)^2 = 7 \cdot 25^k$ and from nothing else, as the merged code shows (`crypto:research/eigen_seta_prototype_20261011/eigen_seta_toy.py`, lines 133–155 and 201–202). Since $k$ is public, anyone can compute the key and decrypt. An exploratory run, not archived, decrypted 5 of 5 toy ciphertexts this way. Version 2 exists to close this hole: the degree of $\tau$ is $2f^2$ with $f$ secret.

### 6.3 Key recovery

**Exhaustive search over $f$.** An attacker can enumerate candidates $f$, apply key generation's public filters, derive a candidate key by Lemma 7, and test it with one decapsulation. The filters accept a fraction $\rho$ of the odd candidates in $[f_{\mathrm{lo}}, 2f_{\mathrm{lo}})$. We measured $\rho$ empirically (`crypto:research/eigen_seta_rust_20261011/reports/run-20261011T1955Z-verify/records/f_filter_rate.txt`): $\log_2\rho = -8.13$, $-8.64$ and $-9.02$ at L1, L3 and L5 (measured from 200, 100 and 100 accepted candidates, so about $\pm 0.1$ to $\pm 0.14$ binomial standard error). This leaves about $2^{267.8}$, $2^{398.6}$ and $2^{525.1}$ admissible degrees, an upper bound since the later rejections of key generation are not counted. The search costs that many chain evaluations classically and their square root with Grover's algorithm. We know of no test that rejects a wrong $f$ more cheaply than a full chain. In the toy experiment of [Ano26b, §6], 85 of 86 wrong but pairing-consistent guesses, including guesses agreeing with the truth on 3–13 low bits, ran the whole chain and failed only at the final split (`crypto:research/eigen_seta_prototype_20261011/results/`).

**Testing a candidate from the public key alone (new in revision 2).** The test above uses a decapsulation, a chain of length $E \approx 4\log_2 f$ on $E_C \times E_C$. A cheaper test needs no ciphertext. Write $\psi_A = 2\tau - tf \in \mathrm{End}(E_A)$. It has trace 0 and degree $7f^2$, and it acts on the public basis as $fs_D$ on $P_A$ and $-fs_D$ on $Q_A$. For a candidate $f$ this action is known exactly. For an odd $u$ and an exponent $e \le E$ with $m = 2^e - 7u^2f^2 = c^2 + c'^2$ ($c$ even), the endomorphism $\gamma = u\psi_A + c$ has degree $7u^2f^2 + c^2 = 2^e - c'^2$. Kani's lemma then gives a $(2^e, 2^e)$-isogeny of $E_A \times E_A$ with kernel $\langle (c'P', -\lambda_S P'), (c'Q', -\lambda_T Q') \rangle$, where $P' = [2^{E-e}]P_A$, $Q' = [2^{E-e}]Q_A$, $\lambda_S = ufs_D + c$ and $\lambda_T = c - ufs_D$.

**Lemma 8** (the public-key test)**.** *For every candidate $f'$ and every plan $(u, e, c, c')$ built from it, the kernel above is isotropic of order $2^{2e}$. The chain splits into a product isomorphic to $E_A \times E_A$ if $f' = f$.*

*Proof.* $e(c'P', c'Q')\, e(\lambda_S P', \lambda_T Q') = e(P', Q')^{c'^2 + \lambda_S\lambda_T}$ and $c'^2 + \lambda_S\lambda_T = c'^2 + c^2 - u^2 f'^2 s_D^2 \equiv c^2 + c'^2 + 7u^2 f'^2 = 2^e$ modulo $2^e$. For $f' = f$ the two generators are $(c'R, -\gamma R)$ for $R = P', Q'$, and $\deg\gamma + c'^2 = 2^e$, so this is Kani's lemma with codomain $E_A^2$ as in Proposition 2. □

The congruence and the norm identity behind the lemma are checked in Lean 4 (`crypto:research/eigen_seta_assessment_20261017/lean/EigenSetaPkTest.lean`). The step from the norm in $\mathbb{Z}[f\sqrt{-7}]$ to the degree of an endomorphism of $E_A$ is not formalised. The chain needs only $e > \log_2(7u^2f^2)$, about $2\log_2 f + 3$, against $E \approx 4\log_2 f$. We implemented the test (`eigenx:src/cryptanalysis/pk_test.rs`) and measured it (§8.4). The true degree is accepted at $e = 569$, 832 and 1087 at L1, L3 and L5. The kernel is isotropic for every candidate, so a wrong one is not rejected early: 158 wrong candidates, including some that agree with $f$ on 16, 64 or $\frac{1}{2}\log_2 f$ low bits, ran to the end and were rejected (none accepted). At L1 one test took 0.79 s against 1.3 s for a decapsulation on the same host (medians). The test therefore cuts the per-candidate cost of the degree search by a factor of about two in chain length, and by 1.6 to 2.0 in measured time at L1 to L5. The cost model now charges this shorter chain (Table 3). The search space is unchanged, and the row stays far above the binding attack.

**Recovering $\varepsilon$ from a path.** Let $I$ be any left $\mathrm{End}(E_D)$-ideal connecting $E_D$ to $E_A$. The isogeny $\varepsilon$ corresponds to an equivalent ideal of norm $f$. There are about $\frac{15}{2\pi^2}X^2$ cyclic isogenies of degree at most $X$ from $E_D$, spread over about $p/12$ curves, so on average about $9X^2/p$ of them land on a given curve. For $X = 2f_{\mathrm{lo}}$ this is about $2^{-782}$ at level 1. So $\varepsilon$ is, with overwhelming probability, the unique shortest isogeny from $E_D$ to $E_A$, and a reduced basis of $I$ yields it, and $f$, in polynomial time. Key recovery therefore costs no more than finding *some* isogeny from $E_D$ to $E_A$. That is the supersingular path or endomorphism-ring problem [EHL+18, Wes22b]: $\tilde{O}(p^{1/2})$ classically [DG16] and $\tilde{O}(p^{1/4})$ quantumly [BJS14]. More recent classical algorithms run in $p^{1/3+o(1)}$ time and memory under a smoothness heuristic [Wes26] and in $p^{2/5+o(1)}$ time without it [Udo26]. The cost model uses $p^{1/3}$. Because $p \approx d^5$, all three are far beyond the target.

**Meet in the middle on $\varepsilon$.** If $f$ is smooth, $\varepsilon$ factors and a meet-in-the-middle search becomes possible. Even with $f$ known and perfectly balanced, it costs at least $\sqrt{f} \geq \sqrt{f_{\mathrm{lo}}}$: $2^{138.5}$, $2^{204.1}$ and $2^{267.5}$ at the three levels, which is above $2^\lambda$, and it applies only to the fraction of keys with smooth $f$.

**Torsion-point attacks on the public key.** The public key reveals $\varepsilon$ on $E_D[2^E]$, with $2^{2E} \gg f$, up to the diagonal scalars $\lambda_1, \lambda_2$. The SIDH attacks [CD23, MMP+23, Rob23] need the degree to build the auxiliary isogeny of degree $2^E - f$ or $N^2 - f$. With $f$ unknown they reduce to the exhaustive search above. Hiding the degree as a countermeasure to these attacks also appears in terSIDH and binSIDH [BF23], where the secret degree ranges over products of small prime powers; here it ranges over an interval. Knowing $f$ would make $\lambda_1, \lambda_2$ irrelevant: $\tau$ acts on $(P_A, Q_A)$ as $f \cdot \mathrm{diag}(\mu, \nu)$ whatever the scaling, and the attacker could simply decrypt.

**Leakage through the public key.** The Weil pairing $e(P_A, Q_A) = e(P_D, Q_D)^{\lambda_1\lambda_2 f}$ is uniform among primitive $2^E$-th roots of unity, because $\lambda_1\lambda_2$ is a uniform unit. The eigenvalues of $\tau$ are $f\mu$ and $f\nu$, with $\mu, \nu$ public, so an attacker knows the action of $\tau$ on $E_A[2^E]$ up to the unknown scalar $f$ mod $2^E$. We do not know how to use this without $f$. The decryption identity $\deg\gamma + c'^2 = 2^E$ involves $f^2$ through $\deg\tau$, so there is no smaller Kani instance that tests $f$ modulo $2^j$: the public-key test of Lemma 8 needs $f$ modulo $2^e$ with $2^e > 7f^2$, which is all of $f$.

This is the main structural assumption of the scheme, and it is where we would look first for an attack.

**The hidden orientation.** $E_A$ carries the orientation by $\mathbb{Z}[f\omega]$, of discriminant $-7f^2$, pushed through $\varepsilon$. The prime 2 splits in $\mathbb{Q}(\sqrt{-7})$ and does not divide the conductor $f$, so the public eigenlines $\langle P_A\rangle$ and $\langle Q_A\rangle$ are kernels of the horizontal ideals $\mathfrak{p}^E$ and $\bar{\mathfrak{p}}^E$. An attacker can therefore walk a segment of length $2E$ of the $\mathfrak{p}$-cycle through $E_A$. Its length, about $4\log_2 f$, is negligible against the order of $[\mathfrak{p}]$ in a class group of size about $f$. The walk does not reach other ideal classes, because evaluating $\mathfrak{l}\star E_A$ for an odd prime $\ell$ needs the action of $\tau$ on $E_A[\ell]$. Every curve $\mathfrak{p}^k\star E_A$ on the segment is again the codomain of a degree-$f$ isogeny from $E_D$, namely the push-forward of $\varepsilon$ through a generator of $\mathfrak{p}_K^k$, since $h(-7)=1$. Given one endomorphism of $E_A$ of discriminant $-7f^2$, $\mathrm{End}(E_A)$ follows in $|-7f^2|^{1/4}\approx 2^{140}$ classical time at L1 and in subexponential quantum time [MW23]. An effective orientation would make the class-group action computable, and Kuperberg's algorithm would then recover the key in subexponential quantum time [Wes22a]. This would hold at every parameter set, the quantum-hardened ones included. No algorithm we know turns the two public walks into an effective orientation. Without one the problem is Wesolowski's $\mathfrak{O}$-Uber, for which only exponential algorithms are known. The level-structure frameworks [FFP24, A$^+$24] are the closest tools we know of, and neither has been instantiated against this public key. The cost model lists $f^{1/2}$ as a classical lower bound for this row (the cost of a generic search in a group of order $f$ if the action could be evaluated); this is in the attacker's favour and is not an attack.

**The form of $\alpha$.** Key generation builds $\alpha = x + y\pi$ with $x, y\in\mathbb{Z}[\omega]$ and $N(y)\le 2^\kappa$, so $y$ ranges over a few thousand values and $x$ is fixed by $(f, y)$ up to units and conjugation. The public curve is determined by $(f, y)$ and a handful of choices, so enumerating $y$ adds only $\kappa$ bits to a search over $f$. We found no use for this structure beyond that. Since it is the only place where the key generator's choices are far from uniform, we record it here.

### 6.4 Message recovery

**Mask guessing plus a torsion attack (binding).** The ciphertext contains $S = a\varphi P_A$ and $T = b\varphi Q_A$, with $\varphi$ of public degree $d$. The pairing $e(S,T) = e(P_A, Q_A)^{abd}$ reveals $ab$, but the ratio $r = b/a$ is uniform. If $r \bmod 2^n$ were known, the attacker would hold $\varphi$ on $E_A[2^n]$ up to the common scalar $a$. Robert's attack in dimension 8 [Rob23] then recovers $\varphi$ in polynomial time as soon as $2^{2n} > d$. With $r$ odd, guessing it costs $2^{n-1}$ trials, where $n = \lfloor(\log_2 d)/2\rfloor + 1$. That is about $2^{128}$, $2^{192}$ and $2^{256}$ dimension-8 chains classically, and the square root of that with Grover. FESTA's analysis [BMP23] rests on the same estimate. An attacker who knew $\tau$ could do better: the masks act diagonally on eigenvectors of $\theta = \varphi\tau\hat{\varphi}$, which is exactly what decapsulation exploits. But knowing $\tau$ means knowing $f$.

**Claw finding on $\varphi$.** Both ends of $\varphi$ are public and its degree is $d$. A generic meet in the middle costs about $\sqrt{d}$ time and memory classically. Tani's algorithm costs $d^{1/3}$ quantum time with as much quantum memory, but under realistic cost models it was not better than classical van Oorschot–Wiener search [vOW99] for SIKE [JS19]. Our $d$ is $2^{256.8}$, $2^{385.1}$ and $2^{513.5}$, which exceeds the degrees $2^{216}$, $2^{305}$ and $2^{372}$ of SIKEp434, p610 and p751 [J$^+$20]. Those were assessed against the same attack for categories 1, 3 and 5. So claw finding is not the binding attack.

**Eigenbases of known endomorphisms.** The Castryck–Vercauteren attacks on FESTA and M-SIDH [CV23] need an endomorphism of $E_A$ of small degree for which $(P_A, Q_A)$ is an eigenbasis. The only such endomorphisms we know are those in $\mathbb{Z}[\tau]$, of degree at least $2f^2$, and they are not available without $f$.

Table 2: Two claim tiers, as $\log_2$. "Queries" is the smallest number of quantum oracle queries over all attacks of Table 3. "Gates" is the smallest gate count under MAXDEPTH $2^{40}/2^{64}/2^{96}$, with the AES reference below it. Derived by `eigenx:examples/security_estimate.rs` at `eigenx` commit `94fe2d9`; record `crypto:research/eigen_seta_assessment_20261017/records/security_estimate.txt`.

| set | $\lambda$ | classical floor | queries | gates (MD $2^{40}/2^{64}/2^{96}$) | tiers met |
|---|---|---|---|---|---|
| L1 | 128 | 128.4 | 64.5 | 151 / 127 / 98 (AES 130 / 106 / 74) | A |
| L3 | 192 | 192.6 | 96.5 | 218 / 194 / 162 (AES 193 / 169 / 137) | A |
| L5 | 256 | 256.8 | 128.5 | 284 / 260 / 228 (AES 258 / 234 / 202) | A (Cat. 5); B at $\lambda = 128$ |
| Q3 | 192 | 384.4 | 192.5 | 415 / 391 / 359 (AES 193 / 169 / 137) | A, B |
| Q5 | 256 | 512.7 | 256.5 | 545 / 521 / 489 (AES 258 / 234 / 202) | A, B |

### 6.5 Quantum security: two claim tiers

Revision 1 of this paper conjectured that L1, L3 and L5 meet NIST categories 1, 3 and 5. That statement conflated two different bars, and we withdraw it in that form. NIST defines the categories by the cost of key search on AES-128, -192 and -256 [Nat16]. Quantumly that cost is stated as a gate count under a depth limit MAXDEPTH $\in \{2^{40}, 2^{64}, 2^{96}\}$: about $2^{170}$, $2^{233}$ and $2^{298}$ gates divided by MAXDEPTH. The binding attack on the L sets is a Grover search over about $2^{\lambda}$ mask ratios, so it needs only about $2^{\lambda/2}$ oracle queries, the same exposure as FESTA [BMP23]. Its oracle, an isogeny chain over a field of about $10\lambda$ bits, is expensive, and Grover parallelises badly under a depth limit. So the gate count can still exceed the AES reference. Table 2 gives both bars, computed by the cost model of the maintained implementation [Ano26a] (`eigenx:src/security.rs`, `eigenx:examples/security_estimate.rs`). The oracle is costed at $2^8$ $\mathbb{F}_{p^2}$ multiplications per $(2,2)$-step with Karatsuba multiplication. That is below any implementation we know, so the gate columns favour the attacker. No external review of that constant exists.

- *Tier A (gate count under MAXDEPTH).* L1, L3 and L5 meet the AES reference at all three depth limits. In this metric they are conjectured to meet categories 1, 3 and 5.
- *Tier B (at least $2^{\lambda}$ quantum queries against every known attack).* Only L5, Q3 and Q5 meet it. The Q sets take $d \geq 2^{4\lambda}$, so the square root of the mask space lands at $2^{\lambda}$. At $\lambda = 128$ this rule reproduces the L5 prime exactly, so L5 is also the quantum-hardened level-1 set ("Q1"). L1 and L3 offer about $2^{64.5}$ and $2^{96.5}$ queries and must not be described as quantum-hardened.

Both tiers assume that the hidden orientation stays ineffective. An effective orientation would bring in Kuperberg's subexponential algorithm and break both tiers at every set.

### 6.6 Chosen-ciphertext security and decryption failures

The KEM is $U^{\not\perp}[\mathsf{Enc}, G]$ [HHK17]. In the random-oracle model it is IND-CCA secure if $\mathsf{Enc}$ is one-way (Assumption 2) and has negligible correctness error $\delta$; quantum random-oracle bounds for implicit rejection also exist [BHH+19]. The adaptive key-recovery attacks that broke static SIDH keys [GPST16] send malformed ciphertexts and watch the result. Here every ciphertext that differs from $\mathsf{Enc}(pk, s)$ for the recovered $s$ yields the pseudorandom $H_R(z, ct)$, in constant time (§7). Decapsulation must also never fail *observably* on malformed inputs. The robustness test (§8) feeds 48 adversarial ciphertexts, including singular and special curves, small-order points and random curves: none causes a panic, and none yields the honest key. Degenerate formula inputs do not give an attacker a failure oracle either. A crafted ciphertext is answered with $H_R(z, ct)$ whether or not an intermediate formula degenerates, because either way the

Table 3: Estimated attack costs, as $\log_2$ of the number of basic operations (revision 2). "Chain" means one isogeny-chain evaluation. All figures are derived from the parameters by the cost model `eigenx:src/security.rs` at commit `94fe2d9`; record `crypto:research/eigen_seta_assessment_20261017/records/security_estimate.txt`. The degree-search rows use the heuristic filter rate $1/\ln 2^E$. The measured rate $\rho$ at L1–L5 (§6.3) is 1.6–2.3 bits larger, so these rows understate the search by that much. The hidden-orientation row is a lower bound under the false assumption that the class-group action can be evaluated. It is not an attack.

| target | attack | L1 | L3 | L5 | Q3 | Q5 |
|---|---|---|---|---|---|---|
| key | degree search, public-key test (chains) | 265.5 | 396.9 | 523.5 | 777.9 | 1035.5 |
| key | same with Grover (queries) | 132.7 | 198.4 | 261.7 | 389.0 | 517.8 |
| key | End($E_A$), classical $p^{1/3}$ | 447.0 | 663.7 | 876.7 | 1301.3 | 1730.7 |
| key | End($E_A$), quantum $p^{1/4}$ | 335.2 | 497.8 | 657.5 | 976.0 | 1298.0 |
| key | hidden orientation, $f^{1/2}$ (lower bound only) | 138.5 | 204.5 | 268.0 | 395.5 | 524.5 |
| message | mask guess + Kani, classical (chains) | 129.0 | 193.0 | 257.0 | 385.0 | 513.0 |
| message | same with Grover (queries) | 64.5 | 96.5 | 128.5 | 192.5 | 256.5 |
| message | claw on $\varphi$, classical $\sqrt{d}$ | 128.4 | 192.6 | 256.8 | 384.4 | 512.7 |
| message | claw on $\varphi$, quantum $d^{1/3}$ | 85.6 | 128.4 | 171.2 | 256.2 | 341.8 |
| target $\lambda$ / tiers met (Table 2) | | 128 / A | 192 / A | 256 / A,$B_{128}$ | 192 / A,B | 256 / A,B |

re-encryption differs. Only the failure of an *honest* ciphertext would be visible, as a mismatch with the encapsulated key. That has probability heuristically $O(E/p)$ (§4), and it did not occur in 3,635 decapsulations. So decryption-failure attacks have nothing to work with. The adaptive attacks on FESTA [MO23] need a validation or timing oracle, which implicit rejection and constant-time decapsulation remove. Mask-leakage hints [DIKM25] and fault injection on torsion masking [GLM26] apply as they do to FESTA. Masks are derived per ciphertext and never reused, and fault resistance is out of scope.

### 6.7 Weak keys

- $\alpha \in \mathbb{Z}[\omega]$, i.e. $y = 0$. The Kani factor would then be an $\mathcal{O}_K$-ideal isogeny, and since $h(-7) = 1$ its codomain is $E_D$ itself (or its conjugate). Key generation forces $(y_0, y_1) \neq (0, 0)$.
- $f \equiv g \pmod{d}$. The pairing test could not tell the two Kani factors apart, so key generation rejects these $f$ via $2f \not\equiv 2^E$.
- Scalars $\equiv -1$ modulo the point order. These break $y$-recovery in the ladder; key generation rejects them.
- Composite or smooth $f$. $f$ is a uniform odd integer prime to 21, so it has small factors with constant probability and is $2^{64}$-smooth with probability about $2^{-9}$ at L1. Attacks that exploit a smooth conductor, as in SCALLOP, walk the class group and so need an effective orientation. Restricting $f$ to primes would cost about 200× more candidates in key generation; it is an optional hardening that we have not applied.
- Self-pairings [CHM+23] need $m^2 \mid \text{disc}$ for an accessible $m$. Here the discriminant $-7f^2$ is odd and prime to 3.
- The schedule condition and the gluing test. These fix the 2-adic shape of $(c', \lambda_S, \lambda_T)$ for all keys. They are public by design, and their cost in key entropy is included in $\rho$.

# 7 Implementation and side channels

Revision 1 described the Rust crate `crypto:eigen_seta/` (about 4,300 lines plus tests), and §7–§8 still describe and measure that historical build unless they say otherwise. The maintained implementation is now the sibling repository `eigenx:` [Ano26a]. §7.5 lists what changed. The crate is generic over the limb count, so one code base serves all parameter sets. Variable-time big integers (`num-bigint`) appear only in key generation and in public constants.

## 7.1 Arithmetic

- $\mathbb{F}_p$. Montgomery multiplication. For our primes $p + 1 \equiv 0 \bmod 2^{64Z}$, where $Z$ is 16, 24 and 32 limbs out of 21, 32 and 42, so the reduction skips the zero limbs of $p + 1$ ("Montgomery-friendly" primes) and costs $N(N - Z)$ instead of $N^2$ word products.
- *Inversion.* Bernstein–Yang safegcd [BY19], generalising the structure of libsecp256k1's `modinv64` to any limb count, with the proven divstep bound. It is 40 to 140 times faster than Fermat inversion at L1–L5 (Table 5).
- $\mathbb{F}_{p^2}$. Karatsuba multiplication, inversion via the norm, and the constant-time square root of Adj and Rodríguez-Henríquez [ARH14].
- *Curves and isogenies.* x-only Montgomery ladders, including a three-point ladder, with Okeya–Sakurai $y$-recovery [OS01]; SIDH-style 3-isogeny chains with optimal strategies [CLN16]; Weierstrass arithmetic with constant-time selection between the doubling and addition formulas; and the gluing, Richelot and splitting maps of [CD23] in Mumford coordinates, with batched inversions. The Weil pairing uses a projective Miller loop over the public bits of $3^F$. Discrete logarithms in $\mu_{3^F}$ are computed digit by digit with constant-time comparisons.

## 7.2 Constant-time decapsulation

Decapsulation is split into two functions so that each can be checked separately.

- `decaps_prepare` depends only on $(pk, ct)$. It parses both, lifts $S$ and $T$, and computes both canonical bases. It runs in variable time.
- `decaps_core` contains everything secret: the Kani chain with its fixed schedule, the identification over the 8 sign and order choices, the pairings and DLPs, the 3-isogeny chain, the re-encryption, and the selection.

In the core, no branch and no memory index depends on secret data, and all loop counts are public. Selections use `subtle`'s constant-time primitives, and every mask passes through `core::hint::black_box`. Scalars have fixed widths. Secret scalars and secret byte buffers are wiped on drop. The only loop whose trip count depends on data, the canonical-basis search, runs on public curves and is capped. A crafted public key or ciphertext makes it return a refusal rather than spin.

**A compiler-introduced leak.** The masks in safegcd's final normalisation (add $m$ if negative, negate if requested) are computed arithmetically, as `r + (m & mask)`. LLVM compiled the add into a conditional that skipped the work when the mask was zero, so the instruction count depended on the sign of a secret-dependent intermediate. That leak affected every field inversion. The source had no branch. The instruction-count probe of §8 found it, and annotating the probe at line level located it. Passing the masks through `black_box` fixed it. This is a reminder that constant-time code must be checked on the binary.

### 7.3 Residual issues

- Key generation is variable time: rejection sampling, Cornacchia and primality tests. It must run where no attacker can time it.
- Allocation: the core allocates vectors whose sizes are fixed by the parameter set. glibc's allocator then executes a number of instructions that depends on the heap state left by earlier variable-time work, though not on secrets. A production implementation should use fixed buffers.
- Points and field elements on the stack are not wiped, only scalars and byte buffers.
- The memory-address side of constant time is argued, not measured: there are no secret-indexed tables. Power, EM and fault attacks are out of scope. The FO check catches faults that corrupt $s$, but not faults that skip the check.
- Constant time is verified for one compiler (rustc 1.97, release profile) and one target (x86-64). Other builds need the probe re-run.

### 7.4 Tests

The crate's tests compare the field arithmetic with a big-integer reference at 1, 2, 9, 21 and 42 limbs, the chain with direct endomorphism evaluation, and the pairing with bilinearity. They also run round trips at Toy16 and Toy32. Known-answer tests pin SHA3-256 digests of the public key, ciphertext and shared secret produced from fixed seeds at Toy16, Toy32 and L1. The robustness tests feed decapsulation 48 adversarial ciphertexts (§6.6) and encapsulation 14 adversarial public keys; every call terminates without a panic. A regression test covers a key-generation hang that we fixed: Tonelli–Shanks accepted $z = 0$ as a non-residue.

### 7.5 The maintained implementation (revision 2)

The following changes in `eigenx:` matter for this paper.

- *Theta-coordinate decapsulation.* The $(2^E, 2^E)$-chain of decapsulation is a level-2 theta chain ported from Theta-Rust (Dartois–Maino–Pope–Robert [DMPR24]; `eigenx:src/theta.rs`, `eigenx:src/chain.rs`). The Mumford chain of §7 is kept for variable-time chain inspection only. The identification of $\gamma$ on $E_C[d]$ now matches $x$-coordinates per image and selects in constant time among the $y$-sign patterns.
- *Parameters.* The Q3 and Q5 sets, the NUMS prime search of (P4), and parallel key-generation attempts. The toy sets are no longer shipped.
- *Cryptanalysis code.* A cost model (`eigenx:src/security.rs`) and runnable attacks (`eigenx:src/cryptanalysis/`): key derivation from $f$, the near-miss experiment, meet in the middle on $\varphi$, mask-guess/Kani scaffolding, and the public-key degree test of Lemma 8 (new in this revision).
- *A field-arithmetic defect, fixed in this revision.* The interleaved products `Fp::mul2` and `Fp::mul3`, which $\mathbb{F}_{p^2}$ multiplication and squaring use, always took the Montgomery-friendly reduction. For a modulus with $p + 1 \not\equiv 0 \bmod 2^{64}$ they returned wrong products. The test `f1` of `eigenx:tests/field.rs`, on the field of $2^{61} - 1$, failed on every run on the main branch and passes after the fix (`crypto:research/eigen_seta_assessment_20261017/records/`). Every Eigen-Séta prime is Montgomery-friendly, so no parameter set was affected. The fix routes other moduli to the generic product, as single multiplication already did.

Constant-time evidence (§8) was produced on the historical Mumford build and has not been repeated on the theta build; `eigenx:SIDE_CHANNELS.md` is the author's assessment of the current code.

# 8 Measurements

§8.4 holds the measurements of revision 2. Everything before it is the historical measurement of revision 1, made on the `crypto:eigen_seta/` build with the Mumford decapsulation chain. Those numbers do not describe the maintained theta build.

All historical measurements were taken on one host: an Intel Xeon at 2.80 GHz (shared cloud VM, 4 logical cores), using rustc 1.97.0 in release mode. Two runs are used. The benchmark run `run-20261011T1813Z-bench` used the code at commit `19c641d12`. The verification run `run-20261011T1955Z-verify` used commit `d504af28e`, which adds the key-generation checks and the buffer wiping. The later commit `72051ac67` only caps the public canonical-basis search; the known-answer tests confirm that it changes no output. Their records are under `crypto:research/eigen_seta_rust_20261011/reports/`.

## 8.1 Sizes and speed

Table 4: Wall time per operation on one thread, as the range over trials (trial count in brackets). Every decapsulation returned the encapsulated key. Record: `run-20261011T1813Z-bench/records/bench_output.txt`.

| set | pk = ct (B) | keygen (s) | encaps (s) | decaps (s) |
|---|---|---|---|---|
| Toy16 | 216 | 0.38 [1] | 0.005–0.008 [5] | 0.12–0.19 [5] |
| Toy32 | 376 | 2.3 [1] | 0.009–0.012 [3] | 0.45–0.59 [3] |
| L1 | 1344 | 32.3 [1] | 0.24–0.26 [3] | 13.4–15.0 [3] |
| L3 | 1992 | 118.6 [1] | 1.03 [2] | 51.8–53.9 [2] |
| L5 | 2632 | 271.8 [1] | 2.17–2.20 [2] | 120.2–124.0 [2] |

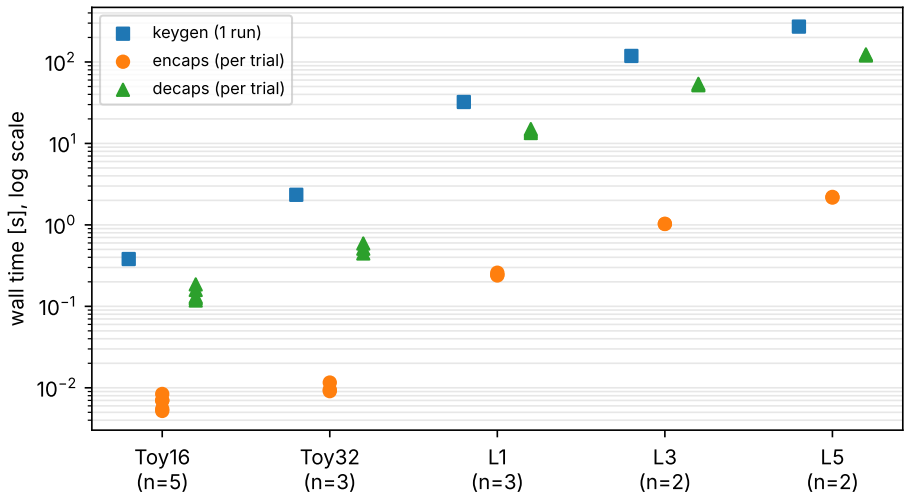



Figure 1: Wall time per operation, every trial shown, log scale. Measured; record as in Table 4.

Encapsulation is fast because it is a 3-isogeny chain of length $F$. Decapsulation is dominated by the $E - 6$ Richelot steps in Mumford coordinates. Each step inverts a polynomial modulo a

quartic for every point it moves; from the formulas, we estimate a few hundred $\mathbb{F}_{p^2}$ multiplications per point and step. Key generation is a single sample, and its time varies widely with the number of rejected candidates.

Table 5: $\mathbb{F}_p$ arithmetic in ns per operation, in the same binary and run. Multiplication is the minimum over 15 repetitions; inversion is the mean of 200. Record: `run-20261011T1813Z-bench/records/bench_output.txt`.

| set | limbs | mul | inversion, Fermat | inversion, safegcd |
|---|---|---|---|---|
| Toy16 | 4 | 54 | 18,403 | 4,741 |
| Toy32 | 6 | 72 | 67,657 | 10,316 |
| L1 | 21 | 611 | 1,426,022 | 35,608 |
| L3 | 32 | 1,516 | 6,708,390 | 109,506 |
| L5 | 42 | 2,449 | 13,453,052 | 96,327 |

No pre-change binary was kept as a baseline for the KEM-level timings, so we claim no KEM-level speedup from the field optimisations.

### 8.2 Correctness

Table 6: Decapsulation outcomes for honest ciphertexts with fresh OsRng keys and ciphertexts. Records: `run-20261011T1955Z-verify/records/failrate/`, plus the benchmark run.

| set | trials | failures | record |
|---|---|---|---|
| Toy16 | 3,005 | 0 | verification run (3,000), benchmark run (5) |
| Toy32 | 603 | 0 | verification run (600), benchmark run (3) |
| L1 | 23 | 0 | verification run (20), benchmark run (3) |
| L3 | 2 | 0 | benchmark run |
| L5 | 2 | 0 | benchmark run |

Zero failures in 3,005 trials bounds the failure rate at Toy16 only by about $10^{-3}$ at 95% confidence (rule of three). The argument that the rate is negligible is the heuristic $O(E/p)$ of §4; the trials show that no systematic failure mode exists.

Beyond these, the known-answer tests and the robustness tests pass: 48 adversarial ciphertexts to decapsulation and 14 adversarial public keys to encapsulation, with no panic and no hang (`crypto:eigen_seta/tests/`).

### 8.3 Constant-time evidence

Callgrind counted the instructions executed by `decaps_core` alone, with keys and ciphertexts generated from fixed seeds (`crypto:eigen_seta/examples/ct_probe.rs`). The count splits into glibc's allocator and `mem*` routines, whose work depends on heap layout but not on secrets (§7), and everything else.

A statistical timing test in the style of dudect [RBV17] (`crypto:eigen_seta/examples/dudect_decaps.rs`) used one Toy16 key and interleaved two classes in random order: a fixed ciphertext and a pool of 16 fresh ones, half of them FO-rejected. With 1,000 samples per class, Welch's statistic was +1.17 on the raw samples and at most 2.36 in absolute value after cropping at the 50th, 75th and 90th percentiles. The usual threshold for a detected leak is $|t| > 4.5$. This test is weaker than the instruction counts: it uses one key, wall-clock time on a shared VM, and compares ciphertext classes rather than keys. Record: `run-20261011T1955Z-verify/records/dudect_output.txt`.

Table 7: Instructions executed in the secret phase of decapsulation, excluding glibc's allocator. "Foreign" ciphertexts are well formed but made for another key, so the FO check rejects them. Records: `run-20261011T1955Z-verify/records/toyprobe/ctprobe_summary.txt` and `.../l1probe/ctprobe_summary.txt`.

| set | keys | ciphertexts per key | non-allocator instructions |
|---|---|---|---|
| Toy16 | 4 | 2 valid, 1 foreign | 1,116,209,216 in all 12 runs |
| L1 | 2 | key 1: 2 valid, 1 foreign; key 2: 1 valid | 120,996,429,610 in all 4 runs |

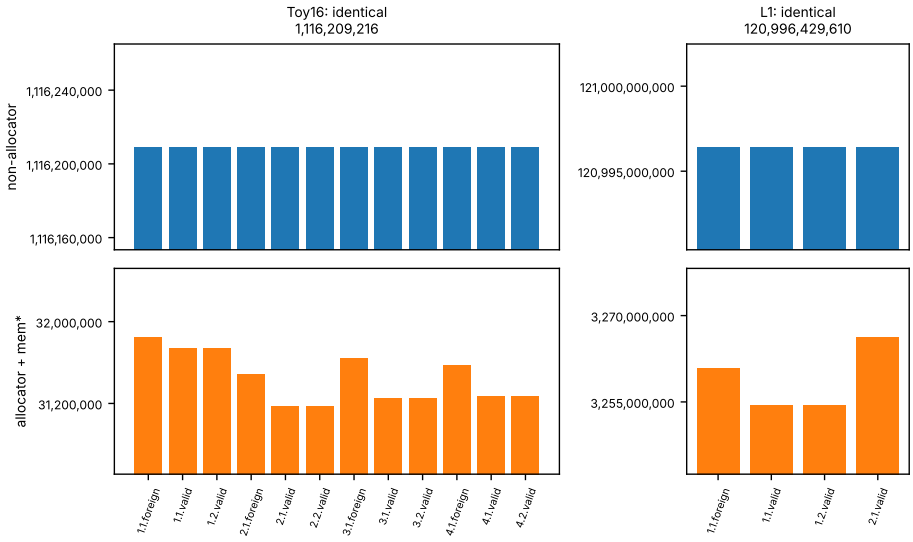



Figure 2: Instruction counts of `decaps_core` per run, at Toy16 (12 runs) and L1 (4 runs). Top: everything except glibc's allocator. Bottom: glibc's allocator and `mem*`. Measured; records as in Table 7.

### 8.4 Revision 2: the public-key degree test

Run `run-20261017T1623Z-pk-degree-test` executed `eigenx:examples/pk_degree_test.rs` at `eigenx` commit `94fe2d9` on a shared cloud VM (Intel Xeon at 2.10 GHz, 4 logical cores, rustc 1.97.0, release build, no core pinning). Records: `crypto:research/eigen_seta_assessment_20261017/records/run-20261017T1623Z-pk-degree-test/`. For each set it generates one key with its true $f$ and tests the true degree and a number of wrong candidates with the theta chain of Lemma 8. A candidate is accepted when the codomain is $E_A$. The wrong candidates are uniform odd values prime to 21 and near misses $f + j2^k$ with $k \in \{16, 64, \lfloor \frac{1}{2} \log_2 f \rfloor\}$, $j \leq 4$. At L1 the true degree and one wrong candidate were also walked with the structural checks of the Mumford inspector. The true degree glued, ran all $e - 6$ Richelot steps and split exactly at the end. The wrong one glued, ran all steps with no early split, and did not split.

The chain is $e/E \approx 0.52$ of the decapsulation chain at every set, because $e \approx 2 \log_2 f$ and $E \approx 4 \log_2 f$. The ratio of the median times is 1.6 at L1 and 2.0 at L3 and L5. Decapsulation also pays for ladders, pairings and the re-encryption. The L1 ranges overlap less cleanly, and we do not explain the lower ratio there. These timings are on a noisy shared VM and support only the factor-of-two reading. The cost model's gate columns for the degree search drop by 1.9 bits as a result. This changes no tier, because the degree search is far from binding.

Table 8: Public-key degree test. $e$ is the chain length for the true degree, $e_{\min} = \lfloor \log_2 7f^2 \rfloor + 2$ is the shortest length the planner tries, and "excess" is the mean of $e - e_{\min}$ over the wrong candidates. The timings are the median and range over interleaved repetitions of one test and one full decapsulation of an honest ciphertext under the same key, wall time with the crate's internal threads. Measured; record as above.

| set | $E$ | $e$ (true $f$) | $e_{\min}$ | excess | wrong (accepted) | test (s) | decaps (s) |
|---|---|---|---|---|---|---|---|
| L1 | 1076 | 569 | 560 | 9.9 | 105 (0) | 0.79 [0.53–0.91], $n$=7 | 1.28 [1.11–1.57] |
| L3 | 1594 | 832 | 823 | 14.7 | 35 (0) | 2.72 [2.64–2.78], $n$=3 | 5.54 [5.49–5.75] |
| L5 | 2106 | 1087 | 1076 | 14.0 | 18 (0) | 6.84 [6.66–7.03], $n$=3 | 13.34 [11.54–14.26] |

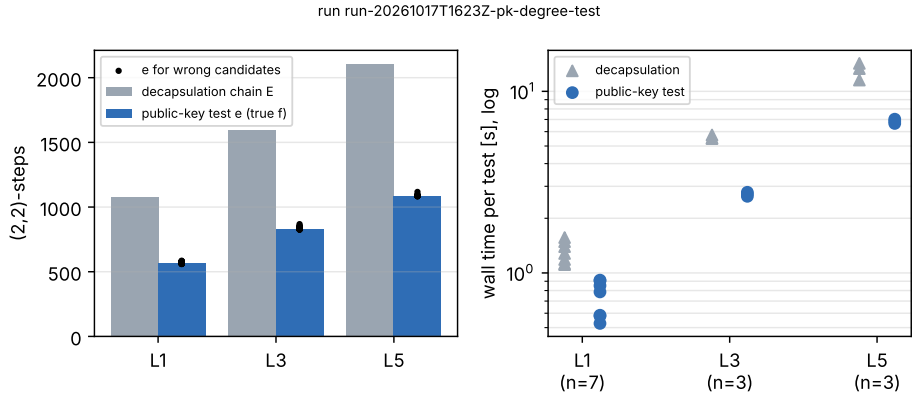



Figure 3: Chain length (left) and wall time per test (right) of the public-key degree test and of decapsulation, every repetition shown. Measured; run `run-20261017T1623Z-pk-degree-test`.

## 9 Assessment (revision 2)

This section records an assessment of the scheme made on 2026-10-17 against the maintained implementation [Ano26a]. It is still the authors' own analysis, not an external review. Its method was as follows. We recomputed every derived number in Tables 1, 3 and 2 from the parameters and the cost model, and reread the proofs of §4. We read the key-generation filter that defines the degree search space. We ran the maintained test suite. We implemented and measured one new attack component, the test of Lemma 8. Records: `crypto:research/eigen_seta_assessment_20261017/`.

**Verdict.** We found no attack below the costs of Table 3. The binding attack is still mask guessing, at about $2^\lambda$ classical chains and $2^{\lambda/2}$ Grover queries on the L sets and $2^\lambda$ queries on L5, Q3 and Q5. Security rests on two assumptions that have no reduction to a standard problem. Assumption 1 is that the hidden-degree eigenbasis with its hidden orientation hides $f$. Assumption 2 is that the masked eigen-torsion problem is hard, and IND-CPA needs its decisional form. We rank the risks as follows.

1. *An effective orientation of $E_A$.* The public key is a segment of length $2E$ of a horizontal 2-walk in the hidden $\mathbb{Z}[f\omega]$-orientation. Any method that extends this to an evaluable class-group action gives subexponential quantum key recovery through Kuperberg's algorithm. It does so at every parameter set, because the Q sets enlarge $d$ and not $f$. The level-structure frameworks [FFP24, A$^+$24] and sesquilinear pairings are the tools to check first. Nobody has done that.

2. *A mask-free higher-dimensional embedding.* The subgroups $\varphi\langle P_A\rangle$ and $\varphi\langle Q_A\rangle$ and the product $ab$ are public and do not depend on the mask ratio. A Kani embedding of

dimension 4 or 8 built from them alone would remove the binding attack's $\sqrt{d}$ search. This is open.

3. *The decisional assumption.* Every basis of $E_A[2^E]$ is an eigenbasis of some endomorphism, so the assumption is about the size of the smallest diagonal one. No distinguisher is known, and no argument for hardness is known beyond the absence of one.

**Corrections to revision 1.**

- *NIST categories* (relabelling, claim correction). Revision 1 conjectured categories 1, 3 and 5 for L1, L3 and L5 on the strength of AES-like structure. That holds only in the gate-count metric (Tier A of §6.5). Against $2^\lambda$ quantum queries (Tier B), L1 and L3 fall short at $2^{64.5}$ and $2^{96.5}$. Only L5, Q3 and Q5 meet that bar.
- *The degree-search oracle* (advance in the attack's cost accounting, small). A candidate degree can be tested on the public key alone with a chain of about $2\log_2 f$ steps instead of a decapsulation of $E \approx 4\log_2 f$ steps (Lemma 8, §8.4). This lowers the degree-search row by about 1.9 bits in the gate model. It changes no tier.
- *Implementation and measurements* (engineering). The paper now points to the maintained code, which decapsulates with a theta chain and adds the Q sets. Revision 1's timings and constant-time evidence describe the historical Mumford build only (§7.5). A field-arithmetic defect for non-friendly moduli was found and fixed. It affected no parameter set.
- *Two inconsistencies in the maintained security notes* (documentation). `eigenx:docs/SECURITY.md` placed the public $\omega$-eigenbasis on $E_D[3^F]$. That is impossible, because 3 is inert in $\mathbb{Q}(\sqrt{-7})$, and Proposition 3 rests on there being no eigenline modulo 3. The note is corrected. Its cost model takes the admissible fraction of degrees to be $1/\ln 2^E$ and says this is the density of sums of two squares. Key generation actually accepts $f$ when $2^E - 7d^2f^2$, after removing prime factors up to $2^{14}$, leaves 1 or a prime $\equiv 1 \pmod 4$, together with the divisibility and sign conditions of Algorithm 1. The measured rate $\rho$ (§6.3) is the right quantity. It exceeds the model's fraction by 1.6 to 2.3 bits, so the model understates the search, which is the conservative direction.

**Practicality.** Primes of about $10.5\lambda$ bits (1341 bits at L1, 3904 and 5192 bits for Q3 and Q5) make public keys and ciphertexts 1.3 to 5.2 kB. Key generation is a rejection search that takes hours at Q5 even with eight workers (`eigenx:research/q5_roundtrip.md`). The decryption chain is the cost driver. Shortening it with two half-chains (open problem 3) would bring $p$ from about $d^5$ to about $d^3$, and it is the most valuable engineering change.

# 10 Open problems

1. *Cryptanalysis of the hidden degree.* The scheme stands or falls with the claim that the public key gives no handle on $f$ beyond exhaustive search (§6.3). In particular, the attacker knows the action of $\tau$ on $E_A[2^E]$ up to the scalar $f \bmod 2^E$. Independent analysis of this point is the most important next step.
2. *Mask entropy.* The binding attack guesses $b/a$ modulo about $\sqrt{d}$. Any way to use fewer torsion bits, for example a dimension-4 variant of the torsion attack, or a lollipop built from another known endomorphism, would lower the security level directly.
3. *Smaller primes.* Decrypting with two half-chains would shrink $2^E$ from about $7d^2f^2$ to its square root and cut $p$ from about $d^5$ to about $d^3$, by the derivation of §5 with $2^{2E} > \deg\gamma$.

4. *The hidden orientation.* Decide whether the two public 2-walks, possibly with level structure [FFP24, A$^+$24] or sesquilinear pairings, make the $\mathbb{Z}[f\omega]$-orientation of $E_A$ effective. A positive answer breaks every parameter set quantumly (§9).

5. *Speed.* Decapsulation now uses theta-coordinate $(2,2)$-chains [DMPR24]. Stack-only buffers and SIMD field arithmetic remain, as does repeating the constant-time evidence on the theta build.

6. *Constant-time key generation*, in particular a constant-time norm-equation solver.

7. *Compression* of public keys and ciphertexts, by encoding the points relative to canonical bases as in compressed SIDH.

## References


[A$^+$24] Sarah Arpin et al. Level-structure group actions on supersingular curves. Cryptology ePrint Archive, Paper 2024/1172; cited as in `eigenx:docs/SECURITY.md`, 2024.

[Ano26a] Anonymous. EigenX: Rust implementation of the Eigen-Séta KEM and PKE, with its security analysis. Sibling repository `eigenx`: `docs/SECURITY.md`, `docs/LITERATURE_DRILL.md`, `src/security.rs`, 2026.

[Ano26b] Anonymous. Public-key encryption from the endomorphism ring problem: research note and toy prototype. Repository files `crypto:RESEARCH_ENDRING_PKE_STRATEGY.md` and `crypto:research/eigen_seta_prototype_20261011/`, 2026.

[ARH14] Gora Adj and Francisco Rodríguez-Henríquez. Square root computation over even extension fields. *IEEE Transactions on Computers*, 63(11):2829–2841, 2014.

[BF23] Andrea Basso and Tako Boris Fouotsa. New SIDH countermeasures for a more efficient key exchange. In *Advances in Cryptology – ASIACRYPT 2023*, volume 14444 of *LNCS*, pages 208–233. Springer, 2023.

[BHH$^+$19] Nina Bindel, Mike Hamburg, Kathrin Hövelmanns, Andreas Hülsing, and Edoardo Persichetti. Tighter proofs of CCA security in the quantum random oracle model. In *Theory of Cryptography – TCC 2019*, volume 11892 of *LNCS*, pages 61–90. Springer, 2019.

[BJS14] Jean-François Biasse, David Jao, and Anirudh Sankar. A quantum algorithm for computing isogenies between supersingular elliptic curves. In *Progress in Cryptology – INDOCRYPT 2014*, volume 8885 of *LNCS*, pages 428–442. Springer, 2014.

[BMP23] Andrea Basso, Luciano Maino, and Giacomo Pope. FESTA: Fast encryption from supersingular torsion attacks. In *Advances in Cryptology – ASIACRYPT 2023*, volume 14444 of *LNCS*, pages 98–126. Springer, 2023.

[BY19] Daniel J. Bernstein and Bo-Yin Yang. Fast constant-time gcd computation and modular inversion. *IACR Transactions on Cryptographic Hardware and Embedded Systems*, 2019(3):340–398, 2019.

[CD23] Wouter Castryck and Thomas Decru. An efficient key recovery attack on SIDH. In *Advances in Cryptology – EUROCRYPT 2023*, volume 14008 of *LNCS*, pages 423–447. Springer, 2023.

[CHM+23] Wouter Castryck, Marc Houben, Simon-Philipp Merz, Marzio Mula, Sam van Buuren, and Frederik Vercauteren. Weak instances of class group action based cryptography via self-pairings. In *Advances in Cryptology – CRYPTO 2023*. Springer, 2023. Cryptology ePrint Archive, Paper 2023/549.

[CLN16] Craig Costello, Patrick Longa, and Michael Naehrig. Efficient algorithms for supersingular isogeny Diffie-Hellman. In *Advances in Cryptology – CRYPTO 2016*, volume 9814 of *LNCS*, pages 572–601. Springer, 2016.

[CV23] Wouter Castryck and Frederik Vercauteren. A polynomial time attack on instances of M-SIDH and FESTA. In *Advances in Cryptology – ASIACRYPT 2023*, volume 14444 of *LNCS*, pages 127–156. Springer, 2023. Cryptology ePrint Archive 2023/1433.

[DG16] Christina Delfs and Steven D. Galbraith. Computing isogenies between supersingular elliptic curves over $\mathbb{F}_p$. *Designs, Codes and Cryptography*, 78(2):425–440, 2016.

[DIKM25] Saikat Das, Riccardo Invernizzi, Péter Kutas, and Jonas Meers. Leveled isogeny problems with hints. PKC 2026; Cryptology ePrint Archive, Paper 2025/2075, 2025.

[DMPR24] Pierrick Dartois, Luciano Maino, Giacomo Pope, and Damien Robert. An algorithmic approach to $(2, 2)$-isogenies in the theta model and applications to isogeny-based cryptography. In *Advances in Cryptology – ASIACRYPT 2024*, LNCS. Springer, 2024. Cryptology ePrint Archive 2023/1747.

[EHL+18] Kirsten Eisenträger, Sean Hallgren, Kristin Lauter, Travis Morrison, and Christophe Petit. Supersingular isogeny graphs and endomorphism rings: Reductions and solutions. In *Advances in Cryptology – EUROCRYPT 2018*, volume 10822 of *LNCS*, pages 329–368. Springer, 2018.

[FFP24] Luca De Feo, Tako Boris Fouotsa, and Lorenz Panny. Isogeny problems with level structure. Cryptology ePrint Archive, Paper 2024/459, 2024.

[FMP23] Tako Boris Fouotsa, Tomoki Moriya, and Christophe Petit. M-SIDH and MD-SIDH: Countering SIDH attacks by masking information. In *Advances in Cryptology – EUROCRYPT 2023*, volume 14008 of *LNCS*, pages 282–309. Springer, 2023.

[GLM26] Valerie Gilchrist, Yi-Fu Lai, and Michael Meyer. Fault injection attacks on torsion masking. Cryptology ePrint Archive, Paper 2026/1944, 2026.

[GPST16] Steven D. Galbraith, Christophe Petit, Barak Shani, and Yan Bo Ti. On the security of supersingular isogeny cryptosystems. In *Advances in Cryptology – ASIACRYPT 2016*, volume 10031 of *LNCS*, pages 63–91. Springer, 2016.

[HHK17] Dennis Hofheinz, Kathrin Hövelmanns, and Eike Kiltz. A modular analysis of the Fujisaki-Okamoto transformation. In *Theory of Cryptography – TCC 2017*, volume 10677 of *LNCS*, pages 341–371. Springer, 2017.

[J+20] David Jao et al. SIKE: Supersingular isogeny key encapsulation. Submission to the NIST Post-Quantum Cryptography Standardization project, round 3, 2020.

[JS19] Samuel Jaques and John M. Schanck. Quantum cryptanalysis in the RAM model: Claw-finding attacks on SIKE. In *Advances in Cryptology – CRYPTO 2019*, volume 11692 of *LNCS*, pages 32–61. Springer, 2019.

[Kan97] Ernst Kani. The number of curves of genus two with elliptic differentials. *Journal für die reine und angewandte Mathematik*, 485:93–121, 1997.

[MMP+23] Luciano Maino, Chloe Martindale, Lorenz Panny, Giacomo Pope, and Benjamin Wesolowski. A direct key recovery attack on SIDH. In *Advances in Cryptology – EUROCRYPT 2023*, volume 14008 of *LNCS*, pages 448–471. Springer, 2023.

[MO23] Tomoki Moriya and Hiroshi Onuki. Adaptive attacks against FESTA without input validation or constant-time implementation. Cryptology ePrint Archive, Paper 2023/1092, 2023.

[MW23] Arthur Herlédan Le Merdy and Benjamin Wesolowski. The supersingular endomorphism ring problem given one endomorphism. IACR Communications in Cryptology 2(1), 2025; Cryptology ePrint Archive, Paper 2023/1448, 2023.

[Nat16] National Institute of Standards and Technology. Submission requirements and evaluation criteria for the post-quantum cryptography standardization process, 2016.

[OS01] Katsuyuki Okeya and Kouichi Sakurai. Efficient elliptic curve cryptosystems from a scalar multiplication algorithm with recovery of the y-coordinate on a Montgomery-form elliptic curve. In *Cryptographic Hardware and Embedded Systems – CHES 2001*, volume 2162 of *LNCS*, pages 126–141. Springer, 2001.

[RBV17] Oscar Reparaz, Josep Balasch, and Ingrid Verbauwhede. Dude, is my code constant time? In *Design, Automation & Test in Europe (DATE 2017)*, pages 1697–1702. IEEE, 2017.

[Rob23] Damien Robert. Breaking SIDH in polynomial time. In *Advances in Cryptology – EUROCRYPT 2023*, volume 14008 of *LNCS*, pages 472–503. Springer, 2023.

[Udo26] Aleksei Udovenko. Solving the supersingular isogeny problem in time $p^{2/5+o(1)}$. Cryptology ePrint Archive, Paper 2026/1575, 2026.

[vOW99] Paul C. van Oorschot and Michael J. Wiener. Parallel collision search with cryptanalytic applications. *Journal of Cryptology*, 12(1):1–28, 1999.

[Wes22a] Benjamin Wesolowski. Orientations and the supersingular endomorphism ring problem. In *Advances in Cryptology – EUROCRYPT 2022*. Springer, 2022. Cryptology ePrint Archive, Paper 2021/1583.

[Wes22b] Benjamin Wesolowski. The supersingular isogeny path and endomorphism ring problems are equivalent. In *62nd IEEE Symposium on Foundations of Computer Science (FOCS 2021)*, pages 1100–1111. IEEE, 2022.

[Wes26] Benjamin Wesolowski. The supersingular isogeny problem in time and memory $p^{1/3+o(1)}$. Cryptology ePrint Archive, Paper 2026/1486, 2026.